Android Application Development Lab

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Q1 -> Develop an application that shows names as a list and on selecting a name it should show the details of the candidate on the next screen with a Back button. If the screen is rotated to landscape mode (width greater than height), then the screen should show list on left fragment and details on right fragment instead of second screen with back button. Use Fragment transactions and Rotation event listener.

Ans1 ->

MainActivity.java

```java
package com.example.masterdetail;

import android.content.Intent;
import android.content.res.Configuration;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.FragmentTransaction;

public class MainActivity extends AppCompatActivity
       implements ListFragment.OnNameSelectedListener {

   private boolean isLandscape;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       isLandscape = getResources().getConfiguration().orientation
               == Configuration.ORIENTATION_LANDSCAPE;

       // Load ListFragment if not already added
       if (savedInstanceState == null) {
           getSupportFragmentManager()
                   .beginTransaction()
                   .add(R.id.listContainer, new ListFragment(), "list")
```

```java
                    .commit();
        }
    }

    @Override
    public void onNameSelected(int index) {
        if (isLandscape) {
            // Landscape: show DetailFragment on the right panel
            DetailFragment detail = DetailFragment.newInstance(index);
            FragmentTransaction ft =
getSupportFragmentManager().beginTransaction();
            ft.replace(R.id.detailContainer, detail, "detail");
            ft.addToBackStack(null);
            ft.commit();
        } else {
            // Portrait: launch DetailActivity
            Intent intent = new Intent(this, DetailActivity.class);
            intent.putExtra("index", index);
            startActivity(intent);
        }
    }
}
```

DetailActivity.java

```java
package com.example.masterdetail;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class DetailActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_detail);

        int index = getIntent().getIntExtra("index", 0);

        if (savedInstanceState == null) {
            DetailFragment fragment = DetailFragment.newInstance(index);
            getSupportFragmentManager()
                    .beginTransaction()
                    .add(R.id.detailContainer, fragment)
                    .commit();
        }
    }
}
```

ListFragment.java

```java
package com.example.masterdetail;

import android.content.Context;
import android.os.Bundle;
import android.view.*;
import android.widget.*;
import androidx.fragment.app.Fragment;

public class ListFragment extends Fragment {

   public interface OnNameSelectedListener {
       void onNameSelected(int index);
   }

   private OnNameSelectedListener listener;

   @Override
   public void onAttach(Context context) {
       super.onAttach(context);
       if (context instanceof OnNameSelectedListener)
           listener = (OnNameSelectedListener) context;
   }

   @Override
   public View onCreateView(LayoutInflater inflater, ViewGroup container,
                            Bundle savedInstanceState) {
       View view = inflater.inflate(R.layout.fragment_list, container,
false);
       ListView listView = view.findViewById(R.id.listView);

       ArrayAdapter<String> adapter = new ArrayAdapter<>(
               getActivity(),
               android.R.layout.simple_list_item_1,
               Candidates.NAMES
       );
       listView.setAdapter(adapter);

       listView.setOnItemClickListener((parent, v, position, id) -> {
           if (listener != null)
               listener.onNameSelected(position);
       });

       return view;
   }
}
```

DetailFragment.java

```java
package com.example.masterdetail;
```

```java
import android.os.Bundle;
import android.view.*;
import android.widget.TextView;
import androidx.fragment.app.Fragment;

public class DetailFragment extends Fragment {

    private static final String ARG_INDEX = "index";

    public static DetailFragment newInstance(int index) {
        DetailFragment f = new DetailFragment();
        Bundle args = new Bundle();
        args.putInt(ARG_INDEX, index);
        f.setArguments(args);
        return f;
    }

    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container,
                             Bundle savedInstanceState) {
        View view = inflater.inflate(R.layout.fragment_detail, container,
false);
        TextView tvDetail = view.findViewById(R.id.tvDetail);

        if (getArguments() != null) {
            int index = getArguments().getInt(ARG_INDEX, 0);
            tvDetail.setText(Candidates.DETAILS[index]);
        }
        return view;
    }
}
```

Candidates.java

```java
package com.example.masterdetail;

public class Candidates {
    public static final String[] NAMES = {
            "Alice Johnson", "Bob Smith", "Carol White",
            "David Brown", "Eva Green"
    };

    public static final String[] DETAILS = {
            "Alice Johnson\nAge: 28\nRole: Android Developer\nExp: 4 years",
            "Bob Smith\nAge: 32\nRole: Backend Engineer\nExp: 7 years",
            "Carol White\nAge: 25\nRole: UI/UX Designer\nExp: 2 years",
            "David Brown\nAge: 30\nRole: ML Engineer\nExp: 5 years",
            "Eva Green\nAge: 27\nRole: DevOps Engineer\nExp: 3 years"
    };
}
```

layout/activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/listContainer"
   android:layout_width="match_parent"
   android:layout_height="match_parent" />
```

land/activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="horizontal"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <FrameLayout
       android:id="@+id/listContainer"
       android:layout_width="0dp"
       android:layout_height="match_parent"
       android:layout_weight="1" />

   <FrameLayout
       android:id="@+id/detailContainer"
       android:layout_width="0dp"
       android:layout_height="match_parent"
       android:layout_weight="2" />

</LinearLayout>
```

activity_detail.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/detailContainer"
   android:layout_width="match_parent"
   android:layout_height="match_parent" />
```

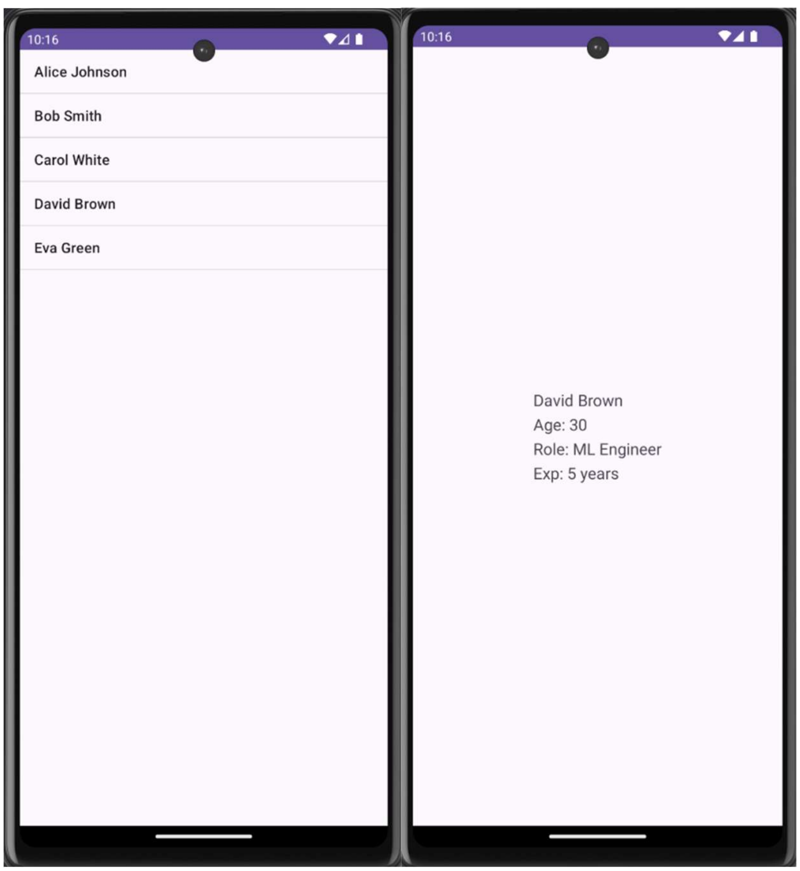
Fragment_detail.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="24dp"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <TextView
       android:id="@+id/tvDetail"
       android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:lineSpacingExtra="6dp"
        android:text="Select a name from the list" />

</LinearLayout>
```

Fragment_list.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```



Q2 ->
i. Create a user registration application that stores the user details in a database table.
ii. Create a database and a user table where the details of login names and passwords are stored. Insert some names and passwords initially. Now the login details entered by the user should be verified with the database and an appropriate dialog should be shown to the user.

Ans2 ->

MainActivity.java
```
package com.example.loginapp;

import android.content.Intent;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   EditText etUsername, etPassword;
   Button btnLogin, btnGoRegister;
   DatabaseHelper db;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       db = new DatabaseHelper(this);

       etUsername    = findViewById(R.id.etUsername);
       etPassword    = findViewById(R.id.etPassword);
       btnLogin      = findViewById(R.id.btnLogin);
       btnGoRegister = findViewById(R.id.btnGoRegister);

       btnLogin.setOnClickListener(v -> {
           String username = etUsername.getText().toString().trim();
           String password = etPassword.getText().toString().trim();

           if (username.isEmpty() || password.isEmpty()) {
               Toast.makeText(this, "Please enter credentials",
Toast.LENGTH_SHORT).show();
               return;
           }
```

```
            boolean valid = db.checkLogin(username, password);

            // Show result as AlertDialog
            AlertDialog.Builder builder = new AlertDialog.Builder(this);
            if (valid) {
                builder.setTitle("Login Successful ✅")
                        .setMessage("Welcome, " + username + "!")
                        .setPositiveButton("OK", null);
            } else {
                builder.setTitle("Login Failed ❌")
                        .setMessage("Invalid username or password.")
                        .setPositiveButton("Try Again", null);
            }
            builder.create().show();
        });

        btnGoRegister.setOnClickListener(v -> {
            startActivity(new Intent(this, RegisterActivity.class));
        });
    }
}
```

RegisterActivity.java

```
package com.example.loginapp;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class RegisterActivity extends AppCompatActivity {

   EditText etUsername, etPassword, etConfirmPassword;
   Button btnRegister;
   DatabaseHelper db;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_register);

       db = new DatabaseHelper(this);

       etUsername = findViewById(R.id.etRegUsername);
       etPassword = findViewById(R.id.etRegPassword);
       etConfirmPassword = findViewById(R.id.etConfirmPassword);
       btnRegister = findViewById(R.id.btnRegister);

       btnRegister.setOnClickListener(v -> {
```

```
            String username = etUsername.getText().toString().trim();
            String password = etPassword.getText().toString().trim();
            String confirm  = etConfirmPassword.getText().toString().trim();

            if (username.isEmpty() || password.isEmpty()) {
                Toast.makeText(this, "Please fill all fields",
Toast.LENGTH_SHORT).show();
                return;
            }
            if (!password.equals(confirm)) {
                Toast.makeText(this, "Passwords do not match",
Toast.LENGTH_SHORT).show();
                return;
            }

            boolean success = db.registerUser(username, password);
            if (success) {
                Toast.makeText(this, "Registered successfully! Please login.",
Toast.LENGTH_SHORT).show();
                finish(); // go back to login screen
            } else {
                Toast.makeText(this, "Username already exists!",
Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

DatabaseHelper.java

```
package com.example.loginapp;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DatabaseHelper extends SQLiteOpenHelper {

   private static final String DB_NAME = "UserDB";
   private static final int DB_VERSION = 1;
   private static final String TABLE_USERS = "users";

   public DatabaseHelper(Context context) {
       super(context, DB_NAME, null, DB_VERSION);
   }

   @Override
   public void onCreate(SQLiteDatabase db) {
```

```
        String createTable = "CREATE TABLE " + TABLE_USERS + " (" +
                "id INTEGER PRIMARY KEY AUTOINCREMENT, " +
                "username TEXT UNIQUE, " +
                "password TEXT)";
        db.execSQL(createTable);

        // Insert some initial users
        db.execSQL("INSERT INTO " + TABLE_USERS + " (username, password)
VALUES ('admin', 'admin123')");
        db.execSQL("INSERT INTO " + TABLE_USERS + " (username, password)
VALUES ('vansh', 'pass123')");
        db.execSQL("INSERT INTO " + TABLE_USERS + " (username, password)
VALUES ('test', 'test123')");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_USERS);
        onCreate(db);
    }

    // Register a new user
    public boolean registerUser(String username, String password) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues values = new ContentValues();
        values.put("username", username);
        values.put("password", password);
        long result = db.insert(TABLE_USERS, null, values);
        return result != -1; // returns false if username already exists
    }

    // Check login credentials
    public boolean checkLogin(String username, String password) {
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor cursor = db.rawQuery(
                "SELECT * FROM " + TABLE_USERS +
                        " WHERE username=? AND password=?",
                new String[]{username, password}
        );
        boolean exists = cursor.getCount() > 0;
        cursor.close();
        return exists;
    }
}
```

Activity_main.xml

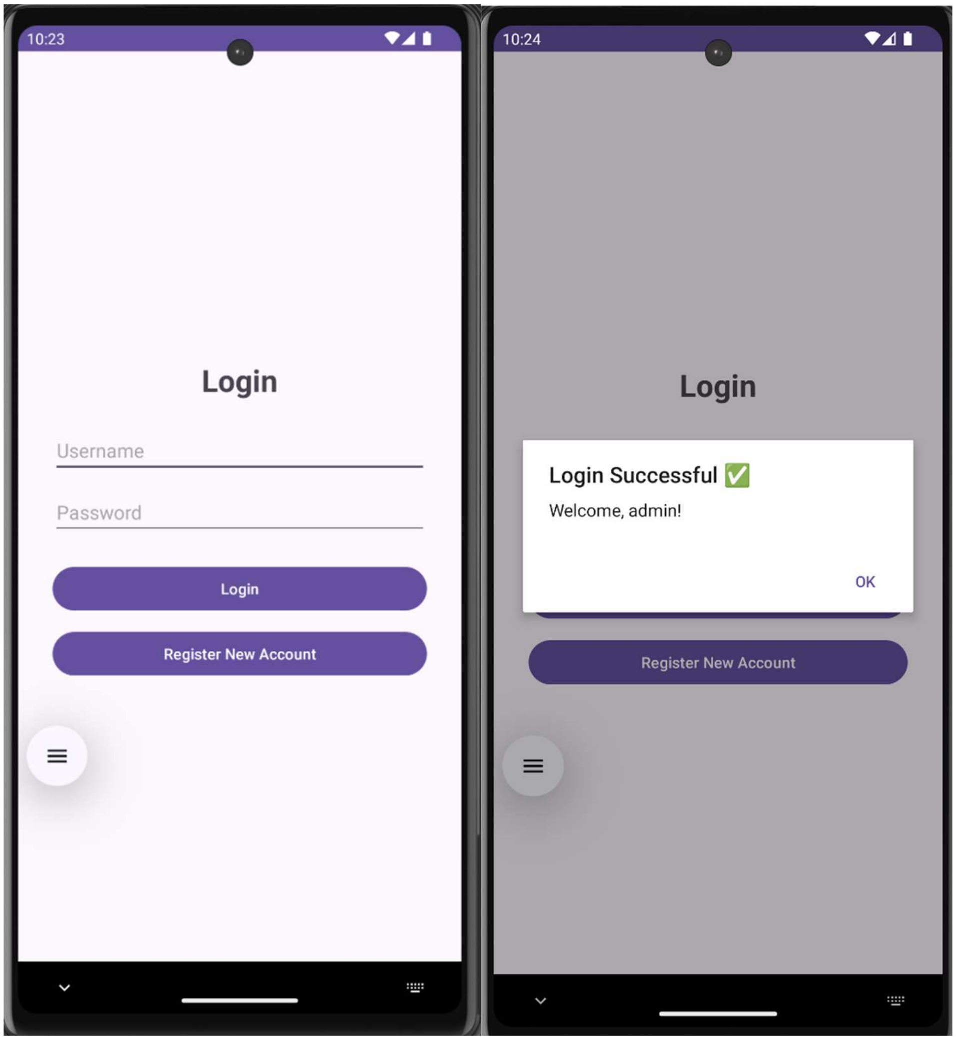
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
```

```xml
    android:gravity="center"
    android:padding="32dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:text="Login"
        android:textSize="28sp"
        android:textStyle="bold"
        android:layout_marginBottom="24dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etUsername"
        android:hint="Username"
        android:inputType="text"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="12dp"/>

    <EditText
        android:id="@+id/etPassword"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="24dp"/>

    <Button
        android:id="@+id/btnLogin"
        android:text="Login"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="12dp"/>

    <Button
        android:id="@+id/btnGoRegister"
        android:text="Register New Account"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

Activity_register.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
```

```xml
    android:padding="32dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:text="Register"
        android:textSize="28sp"
        android:textStyle="bold"
        android:layout_marginBottom="24dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etRegUsername"
        android:hint="Choose Username"
        android:inputType="text"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="12dp"/>

    <EditText
        android:id="@+id/etRegPassword"
        android:hint="Choose Password"
        android:inputType="textPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="12dp"/>

    <EditText
        android:id="@+id/etConfirmPassword"
        android:hint="Confirm Password"
        android:inputType="textPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="24dp"/>

    <Button
        android:id="@+id/btnRegister"
        android:text="Register"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```



Q3 -> Create an admin application for the user table, which shows all records as a list and the admin can select any record for edit or modify. The results should be reflected in the table.

Ans3 ->

MainActivity.java

```
package com.example.useradminapp;

import android.app.AlertDialog;
import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.EditText;
```

```java
import android.widget.LinearLayout;
import android.widget.ListView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import java.util.List;

public class MainActivity extends AppCompatActivity {

   DatabaseHelper db;
   ArrayAdapter<String> adapter;
   List<String> userList;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       db = new DatabaseHelper(this);
       ListView listView = findViewById(R.id.listView);

       loadUsers(listView);

       listView.setOnItemClickListener((parent, view, position, id) -> {
           String[] parts = userList.get(position).split(" \\| ");
           int userId = Integer.parseInt(parts[0]);

           // Build edit dialog
           EditText etName  = new EditText(this);
           EditText etEmail = new EditText(this);
           etName.setText(parts[1]);
           etEmail.setText(parts[2]);
           etName.setHint("Name");
           etEmail.setHint("Email");

           LinearLayout layout = new LinearLayout(this);
           layout.setOrientation(LinearLayout.VERTICAL);
           layout.setPadding(40, 20, 40, 10);
           layout.addView(etName);
           layout.addView(etEmail);

           new AlertDialog.Builder(this)
                   .setTitle("Edit User #" + userId)
                   .setView(layout)
                   .setPositiveButton("Save", (dialog, which) -> {
                       db.updateUser(userId, etName.getText().toString(),
etEmail.getText().toString());
                       Toast.makeText(this, "Updated!",
Toast.LENGTH_SHORT).show();
                       loadUsers(listView); // refresh list
```

```
                    })
                    .setNegativeButton("Cancel", null)
                    .show();
        });
    }

    void loadUsers(ListView listView) {
        userList = db.getAllUsers();
        adapter = new ArrayAdapter<>(this,
android.R.layout.simple_list_item_1, userList);
        listView.setAdapter(adapter);
    }
}
```

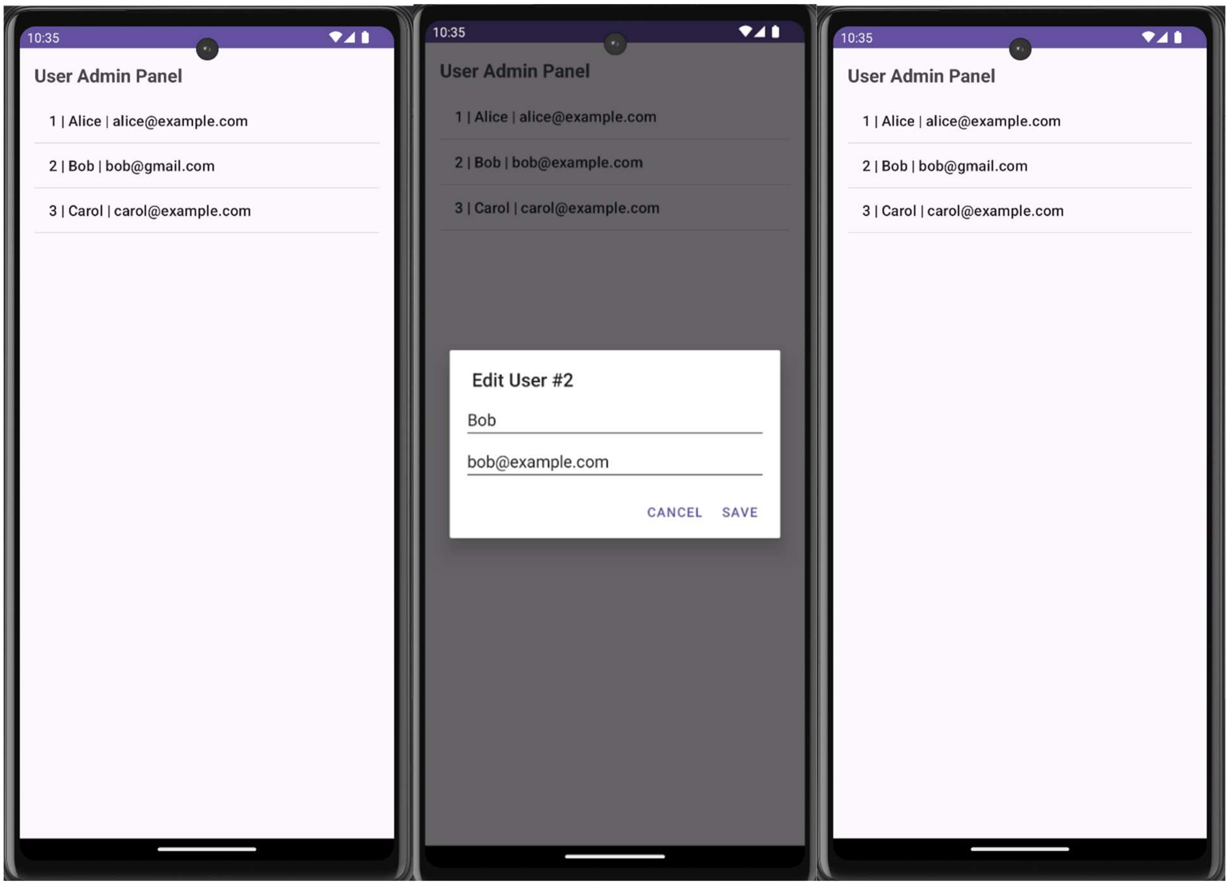
DatabaseHelper.java

```
package com.example.useradminapp;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import java.util.ArrayList;
import java.util.List;

public class DatabaseHelper extends SQLiteOpenHelper {

    public DatabaseHelper(Context context) {
        super(context, "users.db", null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE users (id INTEGER PRIMARY KEY AUTOINCREMENT,
name TEXT, email TEXT)");
        db.execSQL("INSERT INTO users (name, email) VALUES ('Alice',
'alice@example.com')");
        db.execSQL("INSERT INTO users (name, email) VALUES ('Bob',
'bob@example.com')");
        db.execSQL("INSERT INTO users (name, email) VALUES ('Carol',
'carol@example.com')");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int o, int n) {
        db.execSQL("DROP TABLE IF EXISTS users");
        onCreate(db);
    }
```

```java
    public List<String> getAllUsers() {
        List<String> list = new ArrayList<>();
        Cursor c = getReadableDatabase().rawQuery("SELECT * FROM users",
null);
        while (c.moveToNext())
            list.add(c.getInt(0) + " | " + c.getString(1) + " | " +
c.getString(2));
        c.close();
        return list;
    }

    public void updateUser(int id, String name, String email) {
        ContentValues cv = new ContentValues();
        cv.put("name", name);
        cv.put("email", email);
        getWritableDatabase().update("users", cv, "id=?", new
String[]{String.valueOf(id)});
    }
}
```

Activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="User Admin Panel"
        android:textSize="20sp"
        android:textStyle="bold"
        android:layout_marginBottom="12dp"/>

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>

</LinearLayout>
```



Q4 -> Create an alarm that rings every Sunday at 8:00 AM. Modify it to use a time picker to set alarm time.

Ans4 ->

MainActivity.java

```java
package com.example.alarmapp;

import android.app.AlarmManager;
import android.app.PendingIntent;
import android.app.TimePickerDialog;
import android.content.Context;
import android.content.Intent;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;
import java.util.Calendar;

public class MainActivity extends AppCompatActivity {

   Button btnSetSundayAlarm, btnSetCustomAlarm, btnCancelAlarm;
```

```java
    TextView tvStatus;
    AlarmManager alarmManager;
    PendingIntent pendingIntent;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnSetSundayAlarm = findViewById(R.id.btnSundayAlarm);
        btnSetCustomAlarm = findViewById(R.id.btnCustomAlarm);
        btnCancelAlarm    = findViewById(R.id.btnCancelAlarm);
        tvStatus          = findViewById(R.id.tvStatus);

        alarmManager = (AlarmManager) getSystemService(Context.ALARM_SERVICE);

        // Button 1: Set alarm every Sunday at 8:00 AM
        btnSetSundayAlarm.setOnClickListener(v -> {
            Calendar calendar = Calendar.getInstance();
            calendar.set(Calendar.DAY_OF_WEEK, Calendar.SUNDAY);
            calendar.set(Calendar.HOUR_OF_DAY, 8);
            calendar.set(Calendar.MINUTE, 0);
            calendar.set(Calendar.SECOND, 0);

            // If this Sunday has passed, move to next Sunday
            if (calendar.getTimeInMillis() < System.currentTimeMillis()) {
                calendar.add(Calendar.WEEK_OF_YEAR, 1);
            }

            setAlarm(calendar, true);
            tvStatus.setText("✅ Alarm set every Sunday at 8:00 AM");
        });

        // Button 2: Pick custom time using TimePicker
        btnSetCustomAlarm.setOnClickListener(v -> {
            Calendar current = Calendar.getInstance();
            TimePickerDialog timePicker = new TimePickerDialog(
                    this,
                    (view, hourOfDay, minute) -> {
                        Calendar calendar = Calendar.getInstance();
                        calendar.set(Calendar.HOUR_OF_DAY, hourOfDay);
                        calendar.set(Calendar.MINUTE, minute);
                        calendar.set(Calendar.SECOND, 0);

                        // If time has already passed today, set for tomorrow
                        if (calendar.getTimeInMillis() <
System.currentTimeMillis()) {
                            calendar.add(Calendar.DAY_OF_MONTH, 1);
                        }
```

```java
                        setAlarm(calendar, false);
                        tvStatus.setText("✅ Alarm set for "
                                + String.format("%02d:%02d", hourOfDay,
minute));
                    },
                    current.get(Calendar.HOUR_OF_DAY),
                    current.get(Calendar.MINUTE),
                    true // 24-hour format
            );
            timePicker.show();
        });

        // Button 3: Cancel alarm
        btnCancelAlarm.setOnClickListener(v -> {
            Intent intent = new Intent(this, AlarmReceiver.class);
            pendingIntent = PendingIntent.getBroadcast(this, 0, intent,
                    PendingIntent.FLAG_IMMUTABLE);
            alarmManager.cancel(pendingIntent);
            tvStatus.setText("❌ Alarm cancelled");
        });
    }

    private void setAlarm(Calendar calendar, boolean repeating) {
        Intent intent = new Intent(this, AlarmReceiver.class);
        pendingIntent = PendingIntent.getBroadcast(this, 0, intent,
                PendingIntent.FLAG_IMMUTABLE);

        if (repeating) {
            // Repeat every week (7 days in milliseconds)
            alarmManager.setRepeating(
                    AlarmManager.RTC_WAKEUP,
                    calendar.getTimeInMillis(),
                    AlarmManager.INTERVAL_DAY * 7,
                    pendingIntent
            );
        } else {
            alarmManager.setExact(
                    AlarmManager.RTC_WAKEUP,
                    calendar.getTimeInMillis(),
                    pendingIntent
            );
        }
    }
}
```

AlarmReceiver.java

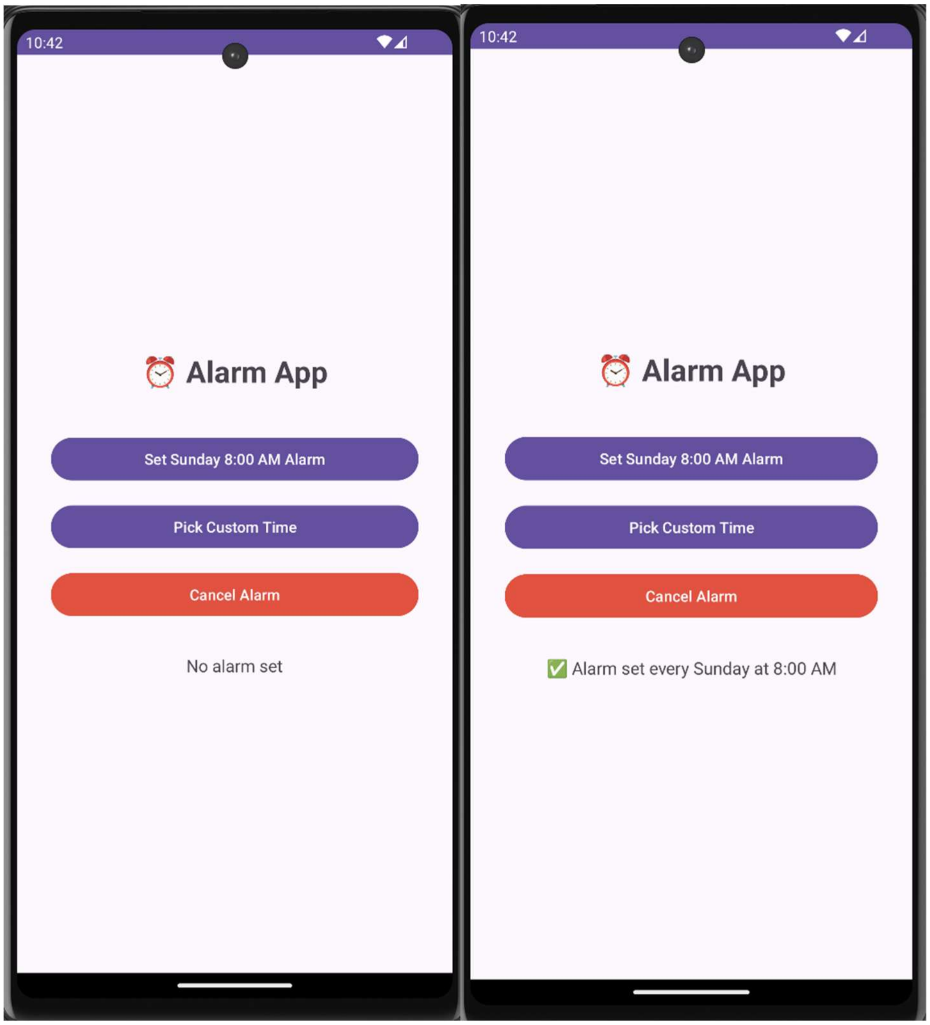
```java
package com.example.alarmapp;
```

```java
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.media.RingtoneManager;
import android.net.Uri;
import android.media.Ringtone;
import android.widget.Toast;

public class AlarmReceiver extends BroadcastReceiver {

   @Override
   public void onReceive(Context context, Intent intent) {
       // Play default ringtone
       Uri alarmUri =
RingtoneManager.getDefaultUri(RingtoneManager.TYPE_ALARM);
       if (alarmUri == null)
           alarmUri =
RingtoneManager.getDefaultUri(RingtoneManager.TYPE_NOTIFICATION);

       Ringtone ringtone = RingtoneManager.getRingtone(context, alarmUri);
       ringtone.play();

       Toast.makeText(context, "⏰ Alarm Ringing!", Toast.LENGTH_LONG).show();
   }
}
```

Activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="32dp"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <TextView
       android:text="⏰ Alarm App"
       android:textSize="28sp"
       android:textStyle="bold"
       android:layout_marginBottom="40dp"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"/>

   <Button
       android:id="@+id/btnSundayAlarm"
       android:text="Set Sunday 8:00 AM Alarm"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
```

```xml
        android:layout_marginBottom="16dp"/>

    <Button
        android:id="@+id/btnCustomAlarm"
        android:text="Pick Custom Time"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="16dp"/>

    <Button
        android:id="@+id/btnCancelAlarm"
        android:text="Cancel Alarm"
        android:backgroundTint="#F44336"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="32dp"/>

    <TextView
        android:id="@+id/tvStatus"
        android:text="No alarm set"
        android:textSize="16sp"
        android:gravity="center"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</LinearLayout>
```



Q5 -> Create an activity that uses a menu with 3 options for dialing a number, opening a website and to send an SMS.

Ans5 ->

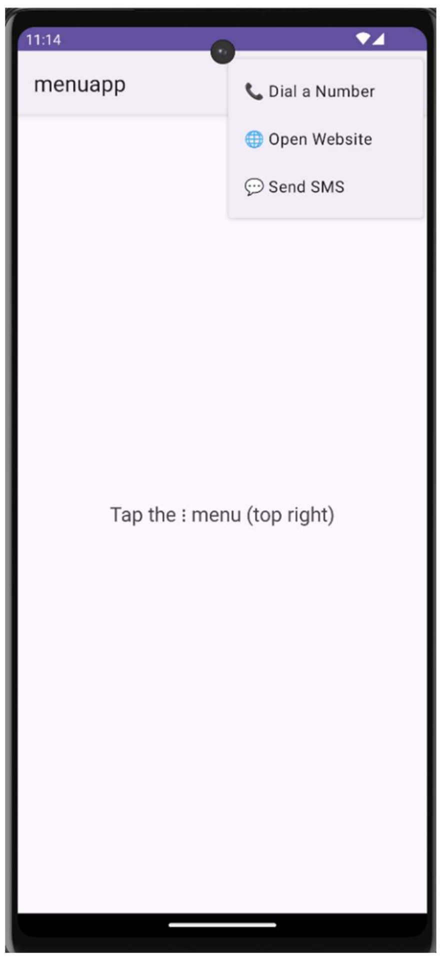
MainActivity.java

```
package com.example.menuapp;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
```

```java
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    // Inflate the menu
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        menu.add(0, 1, 0, "📞 Dial a Number");
        menu.add(0, 2, 0, "🌐 Open Website");
        menu.add(0, 3, 0, "💬 Send SMS");
        return true;
    }

    // Handle menu item clicks
    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        switch (item.getItemId()) {

            case 1: // Dial a number
                Intent dialIntent = new Intent(Intent.ACTION_DIAL);
                dialIntent.setData(Uri.parse("tel:+911234567890"));
                startActivity(dialIntent);
                return true;

            case 2: // Open a website
                Intent webIntent = new Intent(Intent.ACTION_VIEW);
                webIntent.setData(Uri.parse("https://www.google.com"));
                startActivity(webIntent);
                return true;

            case 3: // Send SMS
                Intent smsIntent = new Intent(Intent.ACTION_SENDTO);
                smsIntent.setData(Uri.parse("smsto:+911234567890"));
                smsIntent.putExtra("sms_body", "Hello from the Menu App!");
                startActivity(smsIntent);
                return true;

            default:
                return super.onOptionsItemSelected(item);
        }
    }
}
```

Activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="32dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:text="Tap the ⋮ menu (top right)"
        android:textSize="20sp"
        android:gravity="center"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</LinearLayout>
```



Q6 -> Basic Music Player App

Ans6 ->

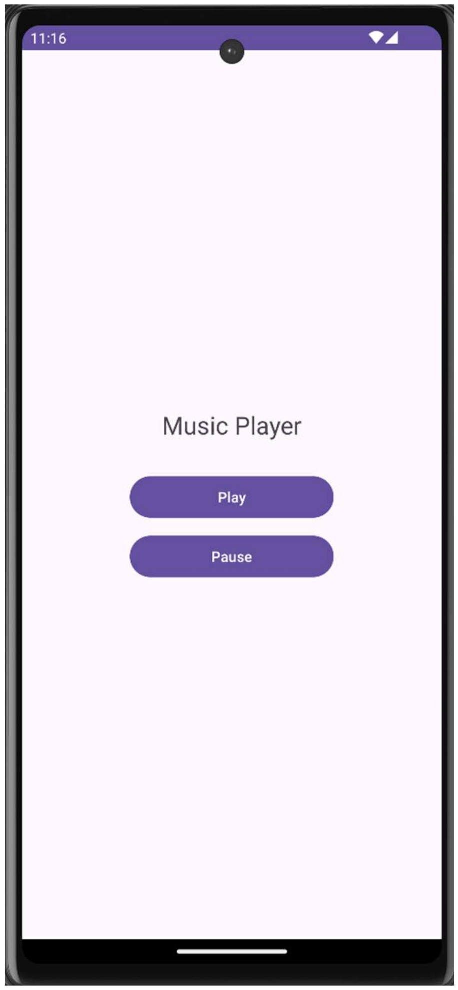
MainActivity.java

```
package com.example.musicplayer;

import android.media.MediaPlayer;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   MediaPlayer mediaPlayer;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       mediaPlayer = MediaPlayer.create(this, R.raw.music_file);

       Button btnPlay = findViewById(R.id.btnPlay);
       Button btnPause = findViewById(R.id.btnPause);

       btnPlay.setOnClickListener(new View.OnClickListener() {
           @Override
           public void onClick(View v) {
               mediaPlayer.start();
           }
       });

       btnPause.setOnClickListener(new View.OnClickListener() {
           @Override
           public void onClick(View v) {
               mediaPlayer.pause();
           }
       });
   }

   @Override
   protected void onStop() {
       super.onStop();
       if (mediaPlayer != null) {
           mediaPlayer.release();
           mediaPlayer = null;
```

```
        }
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">

   <TextView
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Music Player"
       android:textSize="24sp"
       android:layout_marginBottom="30dp"/>

   <Button
       android:id="@+id/btnPlay"
       android:layout_width="200dp"
       android:layout_height="wrap_content"
       android:text="Play" />

   <Button
       android:id="@+id/btnPause"
       android:layout_width="200dp"
       android:layout_height="wrap_content"
       android:text="Pause"
       android:layout_marginTop="10dp"/>

</LinearLayout>
```


11:16
Music Player
Play
Pause